

Project Initialization and Planning Phase

Date	03 JULY 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to improve the analysis and prediction of the Human Development Index (HDI) using Machine Learning. The system integrates socio-economic indicators such as Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI) per capita to provide accurate HDI predictions. The web application enables policymakers, researchers, and students to explore development data, visualize trends, and make informed decisions through real-time predictions and interactive dashboards.

Project Overview	
Objective	The primary objective is to develop an AI-powered system that predicts the Human Development Index (HDI) category using Machine Learning techniques and provides interactive visualizations to support data-driven decision-making.
Scope	The project focuses on analyzing HDI datasets, preprocessing socio-economic indicators, training predictive machine learning models, and deploying the model through a Flask-based web application with interactive dashboard
Problem Statement	
Description	Existing HDI reports mainly provide historical information and require manual analysis of multiple socio-economic indicators. There is no simple, interactive system for instant HDI prediction and visualization.
Impact	Addressing this problem enables policymakers, researchers, students, and organizations to quickly predict development levels, identify improvement areas, and make informed decisions based on accurate data-driven insights.
Proposed Solution	
Approach	Develop a Machine Learning model using HDI indicators, integrate it with a Flask web application, and provide real-time predictions through an intuitive dashboard with graphical visualizations.
Key Features	• Machine Learning-based HDI prediction. • Interactive dashboard with charts and analytics. • Real-time prediction using user inputs. • Dataset exploration and visualization. • Country-wise development comparison. • User-friendly web interface. • Fast and reliable prediction system.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware	Computing Resources	Intel Core i5 / Ryzen 5 or above
	Memory	Minimum 8 GB RAM
	Storage	Minimum 256 GB SSD
Software	Framework	Flask
	Programming Language	Python 3.x
	Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib
	Development environment	Visual code studio, Jupyter Notebook
Data	Dataset	Human Development Index (HDI) Dataset (CSV format) containing Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, GNI per Capita, and HDI category.